Day 147: Time-Series Decomposition: Trend, Seasonality and Residual on Real Data

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 22, Day 1 | Complete study guide

Watch: What is Time Series Decomposition (Aric LaBarr) | Decomposition of Time Series into Trend, Seasonality & Residual from Scratch (Bhavesh Bhatt)

Script: `day147_time_series_decomposition.py` · every number in this guide is copied by program from `day147_run_output.txt`, the real output of that script.

Week 22 turns from images to time series: measurements in time order, where the order matters and the future has to be forecast from the past. Today separates a series into a slow trend, a yearly season and what is left, on two real monthly series: carbon dioxide at Mauna Loa (526 months) and euro-area orders for electrical equipment (257 months), which include the 2008–09 crash. A classical decomposition written from scratch matches statsmodels' exactly (0.0e+00). It shows CO2's rise speeding up (0.86 ppm a year in the 1960s, 1.58 in the 1980s) and its season growing (5.75 to 6.44 ppm), which a single fixed pattern cannot follow. A first forecast made from the decomposition was worse than repeating last year (2.425 against 1.882) until its trend line used ten years instead of two (0.591).

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

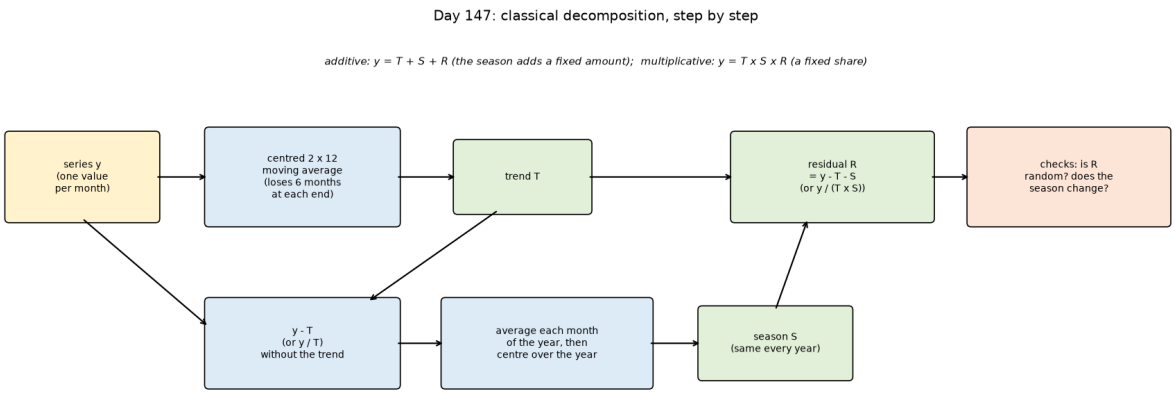
A **time series** is a list of measurements taken in time order, one per month here. **Decomposition** splits it into three parts: the **trend** (the slow, long-run movement), the **season** (a pattern that repeats every year) and the **residual** (what is left after both are taken away).

The week's plan: today, decomposition; Day 148, making a series stationary; Day 149, ARIMA; Day 150, exponential smoothing; Day 151, features for machine-learning models; Day 152, an LSTM; and Day 153 builds a Prophet-style forecaster (a trend plus seasonal terms, fitted together) from the week's parts.

REAL-WORLD SCENARIO

An ice-cream shop's sales fall 40% between August and November. Is the business in trouble? Not if they fall that much every autumn. Decomposition answers the question by separating the usual autumn drop (the season) from the shop's real direction (the trend), so the owner can see whether this autumn is worse than usual (the residual).

SYSTEM DIAGRAM



Classical decomposition: a moving average for the trend, monthly averages for the season, and the rest.

Today's part for the week's build

`day147_exercise.py` is today's parts (the two data loaders, the centred moving average, the classical decomposition, the strength measures and a January-to-December reordering), checked by hand: two moving averages worked on paper, a made-up series with a known trend and season, and the strength of made-up parts.

ACTUAL OUTPUT

```
3-month average of [1.0, 5.0, 2.0, 8.0, 3.0, 7.0, 4.0]: at the second value (1 + 5 + 2) / 3 =
    2.6667; centred_ma gives 2.6667
2 x 4 average at the third value: (0.5 x 1 + 5 + 2 + 8 + 0.5 x 3) / 4 = 4.2500; centred_ma gives
    4.2500
trend 10 + 0.2 t plus a known season, 5 years: largest error in the season 2.0e-15, in the trend
    7.1e-15, largest residual 5.1e-15
strength of made-up parts: trend 1 - 0.01 / 1.16 = 0.99138, season 1 - 0.01 / 1.21 = 0.99174;
    strength() gives 0.99138, 0.99174
CO2: 526 months (5 filled); its season peaks in month 5 (May) at +2.92 ppm and is lowest in month
    10 (October) at -3.14
orders: 257 months, 1995-01 to 2016-05
```

Data: both series ship with statsmodels, so nothing is downloaded. Carbon dioxide (CO2) at Mauna Loa, Hawaii, was recorded weekly; it is averaged into months here, and the few months with no reading at all are filled on a straight line between their neighbours. The orders series is an index of new orders for electrical equipment in the euro area. statsmodels is used only for the data and as a reference to check against; every method below is written in NumPy.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. The data: two real monthly series

DEFINITION — start here, no prior knowledge assumed

Both series have one value per month. CO2 is measured in ppm (parts per million: CO2 molecules per million molecules of air). The orders series is an **index**: a number that only means something compared with other months of the same series.

REAL-WORLD SCENARIO

A thermometer's daily readings and a shop's monthly till receipts are both time series; the first is smooth, the second jumpy. Today's two series are like that: CO2 is smooth, the orders are jumpy and were hit by a recession.

ACTUAL OUTPUT

```
Mauna Loa CO2 (ppm, parts per million; weekly readings averaged by month): 526 months, 1958-03 to 2001-12
  5 months had no reading and were filled on a straight line between neighbours: 1958-06, 1958-10, 1964-02, 1964-03, 1964-04
  lowest 313.4, highest 373.8, first year's mean 315.4, last year's 370.9
euro-area new orders for electrical equipment (an index): 257 months, 1995-01 to 2016-05, none missing
  lowest 65.2 (1995-02), highest 134.1 (2007-12)
Actual output from this lesson's run.
```

What it means

CO2 runs from 1958-03 to 2001-12; 5 months had no reading and were filled. The orders run from 1995-01 to 2016-05 with none missing; their highest month was 134.1 (2007-12), just before the 2008–09 crash.

2. The trend: a centred moving average

DEFINITION — start here, no prior knowledge assumed

A **moving average** replaces each month by the average of the months around it. Averaging over exactly one year cancels the season, because every month of the year is in the window once. With 12 months there is no middle month, so the window takes 13 months and gives the two end months half weight: a **2 × 12** moving average.

REAL-WORLD SCENARIO

To see whether a city is getting warmer, you would not compare one January with the next July; you would compare whole years. A 12-month average does that for every month.

MATHEMATICS

$$T_t = (1/24)\, y_{t-6} + (1/12)\,(y_{t-5} + \ldots + y_{t+5}) + (1/24)\, y_{t+6}$$

y = the series, T = its trend, t = the month; the weights add to 1

IMPLEMENTATION

```
def centred_ma(y, m=M):
    w = np.ones(m) / m if m % 2 else np.r_[0.5, np.ones(m - 1), 0.5] / m
    h = len(w) // 2
    T = np.full(len(y), np.nan)
    T[h:len(y) - h] = np.convolve(y, w, mode="valid")
    return T
```

ACTUAL OUTPUT

```
weights of the 2 x 12 moving average: 0.04167 (= 1/24) at both ends, 0.08333 (= 1/12) for the 11 in between; sum 1.000
a made-up 12-month pattern repeated 6 years: every trend value equals the pattern's mean; largest difference 1.1e-16
a straight line passes through unchanged: largest difference 7.1e-15
so the average removes the season and keeps a straight trend; it loses 6 months at each end
CO2's trend, average rise per year in each decade:
  1960s: 0.86 ppm per year (316.6 -> 325.2)
  1970s: 1.26 ppm per year (325.2 -> 337.8)
  1980s: 1.58 ppm per year (337.8 -> 353.5)
  1990s: 1.51 ppm per year (353.5 -> 368.7)
Actual output from this lesson's run.
```

What it means

Two checks by construction: a repeating 12-month pattern comes out as its own average (largest difference 1.1e-16), and a straight line comes out unchanged (7.1e-15). So the average removes the season and keeps a straight trend. The price is 6 months lost at each end.

CO2's trend rose faster from decade to decade until the 1980s: 0.86 ppm a year in the 1960s, 1.26 in the 1970s, 1.58 in the 1980s and 1.51 in the 1990s, about the 1980s' pace. The trend is not a straight line, which matters for forecasting (section 6).

WHY CHOOSE THIS?

Simple, with no settings to tune beyond the season's length, and exact for a straight trend plus a fixed season.

WHY NOT THIS? (TRADEOFFS)

It loses the last six months, exactly the ones a forecast starts from, and it bends around shocks such as the 2008 crash (section 5).

3. Classical decomposition, checked against statsmodels

DEFINITION — start here, no prior knowledge assumed

After the trend is removed, what is left is mostly season. The classical method averages it for each month of the year (all the Januaries, all the Februaries, ...) and shifts the 12 averages so they add to zero over a year. Whatever the trend and season do not explain is the residual.

The **strength** of a part measures how small the residual is beside it: near 1, the part is clear; near 0, it is lost in the noise.

REAL-WORLD SCENARIO

A teacher who wants each pupil's 'usual' mark for each subject averages that pupil's marks in maths across the year, in English across the year, and so on. The season is the same idea with months instead of subjects.

MATHEMATICS

S_k = *average of* $(y_t - T_t)$ *over every month* t *in position* k*, minus the mean of the 12 averages*

$R_t = y_t - T_t - S_t$

strength of trend $= \max(0, 1 - Var(R) / Var(T + R))$; *of season* $= \max(0, 1 - Var(R) / Var(S + R))$

IMPLEMENTATION

```
def classical_decompose(y, m=M, kind="additive"):
    T = centred_ma(y, m)
    D = y - T if kind == "additive" else y / T
    pos = np.arange(len(y)) % m
    S0 = np.array([np.nanmean(D[pos == k]) for k in range(m)])
    S0 = S0 - S0.mean() if kind == "additive" else S0 / S0.mean()
    S = S0[pos]
    R = y - T - S if kind == "additive" else y / (T * S)
    return T, S, R
```

ACTUAL OUTPUT

```
largest difference from statsmodels' seasonal_decompose (period 12):
     CO2 additive       trend 0.0e+00  seasonal 0.0e+00  residual 0.0e+00
     CO2 multiplicative trend 0.0e+00  seasonal 0.0e+00  residual 2.2e-16
  orders additive       trend 0.0e+00  seasonal 0.0e+00  residual 0.0e+00
  orders multiplicative trend 0.0e+00  seasonal 0.0e+00  residual 2.2e-16
CO2, additive seasonal pattern by calendar month:
      Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct    Nov    Dec
    -0.02   0.63   1.42   2.50   2.92   2.28   0.74  -1.32  -3.04  -3.14  -2.05  -0.92
   peak May, low Oct, swing 6.07; residual std 0.283; strength of trend 1.000, of season 0.981
orders, additive seasonal pattern by calendar month:
      Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct    Nov    Dec
    -9.12  -9.23   7.16  -5.32  -6.83   6.68  -2.94 -14.30  12.07   4.15   7.73   9.95
   peak Sep, low Aug, swing 26.37; residual std 2.661; strength of trend 0.956, of season 0.912
strength: 1 - var(residual) / var(component + residual); near 1 means the residual is small beside that component
Actual output from this lesson's run.
```

What it means

The from-scratch version agrees with statsmodels' `seasonal_decompose` to the last digit (0.0e+00, or 2.2e-16 for the multiplicative residual), for both series.

CO2 peaks in May and is lowest in Oct, a swing of 6.07 ppm: plants in the Northern Hemisphere take up CO2 as they grow in summer, the usual explanation. Its trend and season are both strong (1.000 and 0.981). The orders swing 26.37 index points, peak in Sep and drop in Aug (a likely reason, not tested here: Europe's summer break); their strengths are lower (0.956 and 0.912) because the series is noisier (residual 2.661).

WHY CHOOSE THIS?

Every step can be done by hand, it matches the standard library exactly, and the strength measures say how much of the series each part explains.

WHY NOT THIS? (TRADEOFFS)

It assumes the season is the same every year and that the trend is smooth; sections 4 and 5 test both assumptions.

4. Additive or multiplicative?

DEFINITION — start here, no prior knowledge assumed

In an **additive** decomposition the season adds the same amount every year (y = T + S + R). In a **multiplicative** one it adds the same *share* (y = T × S × R): when the level doubles, so does the seasonal swing. The test is whether the yearly swing grows with the level, and which fits the data better.

REAL-WORLD SCENARIO

A shop whose December sales are always £5,000 above November is additive; a shop whose December sales are always 20% above November is multiplicative, and its December jump grows as the shop grows.

MATHEMATICS

multiplicative: S_k *= average of* y_t / T_t *for month position k, divided by the mean of the 12 averages*

fit error = square root of the mean of $(y - \text{fitted})^2$*, fitted =* $T + S$ *or* $T \times S$*, in the series' own units*

ACTUAL OUTPUT

```
additive: y = T + S + R (the season adds the same amount every year); multiplicative: y = T x S x R (the same share)
  series  corr(swing, level)  fit error: additive  multiplicative
     CO2               0.616               0.2830          0.2575
  orders               0.703               2.6607          2.4701
fit error: root mean squared difference between the series and trend + season (or trend x season), in its own units
orders, multiplicative seasonal factors by calendar month (1.000 = an ordinary month):
      Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct    Nov    Dec
    0.909  0.908  1.070  0.948  0.932  1.065  0.969  0.861  1.122  1.040  1.075  1.100
Actual output from this lesson's run.
```

What it means

In both series the swing grows with the level (correlation 0.616 for CO2, 0.703 for the orders), and the multiplicative fit is closer: 0.2575 against 0.2830 ppm for CO2, 2.4701 against 2.6607 for the orders. The difference is small for CO2, whose yearly mean rose only from 315.4 to 370.9 ppm.

As shares, the orders run 0.861 of an ordinary month in August and 1.122 in September. Sections 5 and 6 use the multiplicative form for the orders.

WHY CHOOSE THIS?

Multiplicative suits series whose swings scale with their level, as most sales and orders do.

WHY NOT THIS? (TRADEOFFS)

It cannot handle zeros or negative values, and an additive model on the logarithm of the series is often the simpler way to get the same effect (section 5 does this for STL).

5. What a classical decomposition gets wrong

DEFINITION — start here, no prior knowledge assumed

Three weaknesses. The **ends**: the moving average has no value for the first and last six months. A **changing season**: one pattern is used for every year. A **shock**: a sudden fall is smoothed into the trend and spills into the residual.

STL (Seasonal-Trend decomposition using Loess, a local smoothing method) is the usual modern alternative: it lets the season change slowly, and its robust option stops unusual months from bending the fit. It is used here from statsmodels, as a reference, not rebuilt.

REAL-WORLD SCENARIO

A recipe for 'the usual Sunday lunch' written in 1960 and never updated will be slightly wrong today; an STL season is a recipe that is quietly revised every year.

SPECIFICATION

Changing season	CO2's classical season estimated separately on 1959–1968 and on 1992–2001
Leftover structure	autocorrelation of the residual one month apart and twelve months apart (0 for pure noise), classical against STL (period 12, robust)
Shock	the orders, multiplicative: the trend through 2008–09, the largest residuals, and the season re-estimated without 2008–2010

ACTUAL OUTPUT

```
ends: no trend for the first 6 and last 6 months (CO2: before 1958-09 and after 2001-06); the latest months are the ones a
    forecast needs
CO2 season from 1959-1968 alone: peak May +2.77, low Oct -2.98, swing 5.75
CO2 season from 1992-2001 alone: peak May +2.99, low Sep -3.45, swing 6.44
one fixed pattern for all 44 years cannot follow that; what it misses stays in the residual, which is not random noise:
  residual autocorrelation, lag 1 and lag 12: classical 0.403, 0.264; STL 0.393, -0.129
  STL's season, averaged over the first and last 10 years: swing 4.56 -> 6.70
the 2008-09 crash in the orders:
  trend peak 122.9 (2007-11), lowest after it 91.5 (2009-08)
  five largest residuals: 1995-09 +8.4%, 1996-09 +7.0%, 2008-10 +7.1%, 2009-01 -7.7%, 2009-02 -8.4%
  seasonal factors with 2008-2010 left out: largest change in any month 0.007
  residual std (as a share): classical 0.0255; STL on the log 0.0303
Actual output from this lesson's run.
```

What it means

CO2's season grew: a swing of 5.75 ppm in 1959–1968 and 6.44 in 1992–2001, with the low moving from October to Sep. STL's season shows the same growth (4.56 -> 6.70). A single pattern cannot follow it, and the classical residual is not noise: months twelve apart are correlated (0.264). STL removes that (-0.129), but neither removes the month-to-month correlation (0.403 and 0.393). That leftover short-term memory is what ARIMA (Day 149) models.

The crash: the orders' trend fell from 122.9 (2007-11) to 91.5 (2009-08). The moving average begins to fall as soon as the crash enters its window, because each month's value uses six months on either side. The five largest residuals are three months of the crash (such as 2009-02 -8.4%) and the Septembers of 1995 and 1996, and leaving 2008–2010 out changes no monthly factor by more than 0.007: the season survived the

crash. STL on the log had the larger residual (0.0303 against 0.0255); a likely reason, not tested here, is that its robust option deliberately leaves unusual months in the residual instead of bending the trend to them.

WHY CHOOSE THIS?

Knowing the weaknesses tells you when the simple method is enough (a stable season, no shocks) and when to use STL.

WHY NOT THIS? (TRADEOFFS)

STL has more settings (the smoothing windows) and, used here from a library, is a black box until rebuilt.

6. Using it: seasonal adjustment, and a first forecast

DEFINITION — start here, no prior knowledge assumed

Seasonal adjustment divides out (or subtracts) the season, so that one month can be compared fairly with the next: it answers 'was this month unusual, for this time of year?'

A **forecast** from a decomposition extends the trend and adds the season back. It is compared with two simple **baselines**: repeating the last value, and repeating the same month of last year.

REAL-WORLD SCENARIO

Unemployment figures in the news are 'seasonally adjusted' so that a rise in January, when temporary Christmas jobs end, is not reported as a crisis every year.

MATHEMATICS

$$\textit{seasonally adjusted} = y_t / S_t \ \textit{(multiplicative)}$$

$$\textit{forecast}_t = a + b\,t + S_{\textit{month of } t},\ \textit{with } a + b\,t \textit{ a straight line through the last } k \textit{ trend values}$$

IMPLEMENTATION

```
for k in (24, 60, 120):
    ok = valid[-k:]; b, a = np.polyfit(ok, T[ok], 1)
    dec[k] = b * steps + a + S[:12][pos]
    print(f"  {'trend line (last ' + str(k) + ' months) + season':<37} {np.mean(np.abs(dec[k] -
        yc[n:])):6.3f} {np.mean(dec[k] - yc[n:]):+7.3f}"
          f"   slope {12 * b:.2f} ppm per year")
```

ACTUAL OUTPUT

```
orders, month-on-month changes: raw and seasonally adjusted point opposite ways in 0.414 of months
  2009-01: raw -27.7%, seasonally adjusted -12.5%
  2006-09: raw +37.0%, seasonally adjusted +5.0%
  2008-11: raw +0.7%, seasonally adjusted -2.7%
CO2 forecast for 2000-01 to 2001-12 (24 months) from data up to 1999-12 (training trend ends 1999-06);
training trend's rise over each of its last three years: 1996-06 -> 1997-06: +1.10; 1997-06 -> 1998-06: +2.88; 1998-06 ->
    1999-06: +1.68
mean absolute error and mean error (forecast - actual), ppm:
  last value                          2.420  -2.207
  same month last year                1.882  -1.882
  trend line (last 24 months) + season   2.425  +2.425   slope 2.48 ppm per year
  trend line (last 60 months) + season   0.890  +0.890   slope 1.88 ppm per year
  trend line (last 120 months) + season  0.591  -0.591   slope 1.52 ppm per year
the line's window is a choice; picking it by this table would use the test months to choose it (later this week:
choosing on earlier data only)
Actual output from this lesson's run.
```

What it means

Raw and adjusted month-to-month changes point opposite ways in 0.414 of months: much of the raw movement is just the calendar. The biggest raw rise, +37.0% in September 2006, was +5.0% after adjustment. January 2009's -27.7% was still -12.5% adjusted: a real collapse, not just January.

The first forecast was a failure: a line through the last 24 trend values climbed 2.48 ppm a year and overshot every month, mean absolute error 2.425 ppm, all of it bias (+2.425), worse than repeating last year (1.882). The training trend's last two years include 1997-06 -> 1998-06: +2.88 ppm, well above the years either side (+1.10 and +1.68). With 60 months the error fell to 0.890, with 120 months to 0.591.

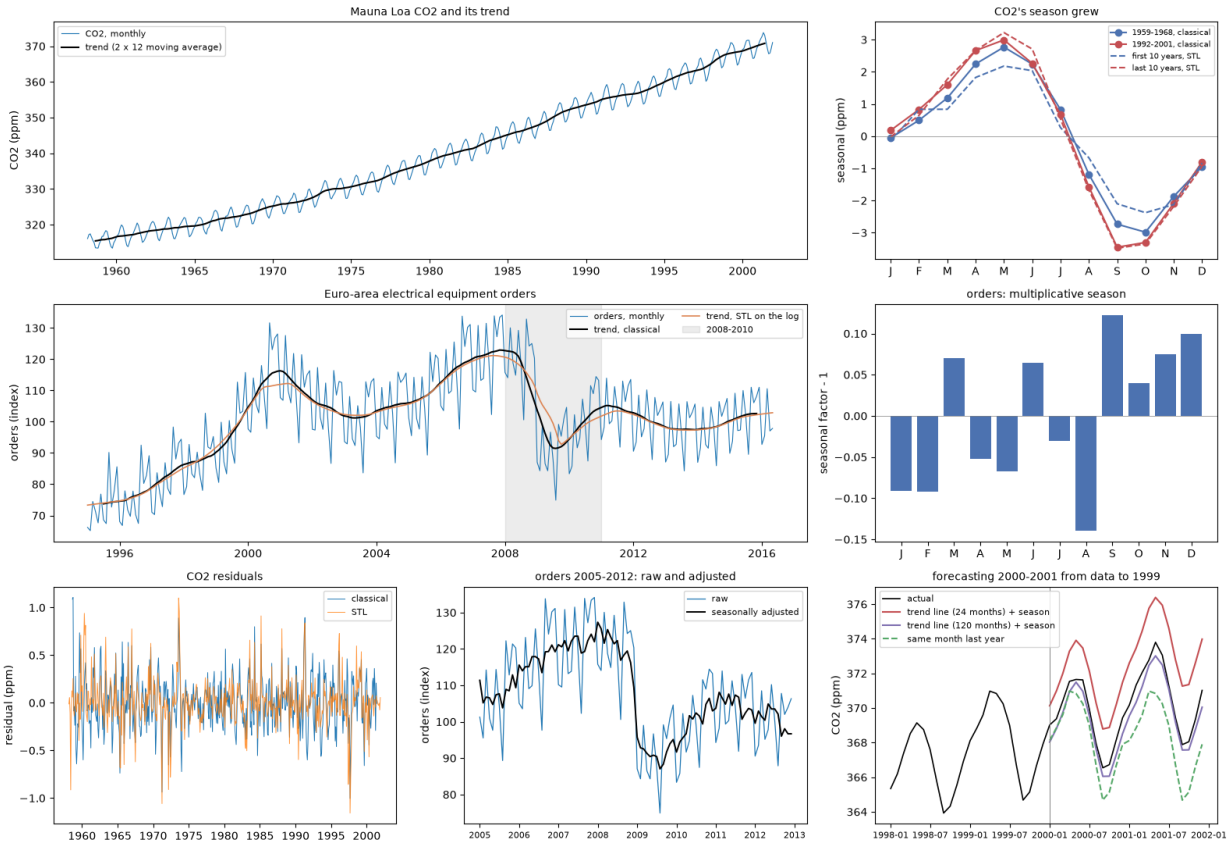
This table must not be used to pick the window: that would choose with the very months being forecast. Choosing a setting on earlier data only, then testing on later data, comes later this week.

WHY CHOOSE THIS?

Seasonal adjustment makes month-to-month comparisons honest, and a decomposition forecast is easy to explain.

WHY NOT THIS? (TRADEOFFS)

A straight-line trend is fragile: its slope depends heavily on how much history it is fitted to, and the classical trend stops six months before the data does.



Top: CO2 and its trend; CO2's season by calendar month, classical for each decade and STL for the first and last ten years. Middle: the orders with the classical and STL trends (2008–2010 shaded), and their multiplicative season. Bottom: CO2 residuals, classical and STL; the orders raw and seasonally adjusted, 2005–2012; CO2 forecasts for 2000–2001.

Interview preparation — technical questions

Q: Why a 2 x 12 moving average rather than a 12-month one?

A: With an even window there is no middle month; the 2 x 12 version centres the window on a month by giving the two end months half weight. It removes a fixed 12-month pattern exactly (1.1e-16 here).

Q: How do you choose between additive and multiplicative?

A: Check whether the seasonal swing grows with the level and compare the fit errors in the series' own units; here multiplicative won for both (2.4701 against 2.6607 for the orders).

Q: What is seasonal adjustment for?

A: Comparing a month with the one before without the calendar's effect; here raw and adjusted changes pointed opposite ways in 0.414 of months.

Q: Why might a decomposition forecast lose to a seasonal naive forecast?

A: A straight trend fitted to a short window can catch a temporary burst; here a 24-month line overshot by 2.425 ppm against 1.882 for repeating last year.

Q: What should a good residual look like?

A: Like noise: no correlation between nearby months. Here the classical residual had 0.403 one month apart, so structure remained.

Interview preparation — theoretical questions

Q: Why does the moving average lose months at the ends?

A: It needs six months on each side of a month; the first and last six have no full window. Methods such as STL fill the ends with local fits instead.

Q: Why did CO2's seasonal swing grow?

A: Measured: 5.75 to 6.44 ppm between the 1960s and 1990s. The cause was not tested here; studies of Mauna Loa attribute it to stronger plant growth and decay in northern ecosystems.

Q: Is the classical decomposition still useful when STL exists?

A: Yes as a transparent baseline that can be checked by hand and matches the library exactly; STL is better when the season drifts (lag-12 residual correlation 0.264 classical, -0.129 STL).

Q: Why not choose the trend window using the test months?

A: Because then the test is no longer a test: the 120-month window looks best (0.591) only after seeing the answers.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> A series can be split into a trend, a season and a residual with nothing more than averages, and the from-scratch version matches the standard library exactly. The split is also a diagnosis: CO2's trend accelerates, its season grows, the orders' season survived a crash that reshaped their trend, and the residual still has memory. Each of those is a reason for a method later this week.

- Classical decomposition from scratch matches statsmodels (0.0e+00).
- CO2's trend: 0.86 ppm a year in the 1960s, 1.58 in the 1980s; its season grew 5.75 to 6.44.
- Multiplicative fitted better for both series (orders 2.4701 against 2.6607).
- The residual is not noise: correlation one month apart 0.403 (classical), 0.393 (STL).
- Forecast: trend line over 24 months 2.425, same month last year 1.882, line over 120 months 0.591.

Suggested watch order, if starting from zero

- 1. What is Time Series Decomposition (Aric LaBarr) — what the three parts are and why a series is split into them.
- 2. Decomposition of Time Series into Trend, Seasonality & Residual from Scratch (Bhavesh Bhatt) — the same decomposition coded from scratch in Python, to compare with today's.

Day 148 preview

- **Stationarity and differencing.** Many forecasting models need a series whose behaviour does not drift. Differencing (each month minus the one before, or minus the same month last year) removes trend and season, and the augmented Dickey-Fuller test, built from scratch, checks whether it worked.